

Creating RDS using Terraform

To Begin with the Lab

Summary of the Lab

This set of labs demonstrated infrastructure automation on AWS using Terraform. It covered provisioning an RDS MariaDB instance and connecting to it from EC2, creating IAM users and roles with appropriate permissions, enabling secure S3 access through IAM roles, configuring EC2 Auto Scaling with CloudWatch alarms to automatically scale resources based on CPU utilization, and implementing an Elastic Load Balancer to distribute traffic across EC2 instances. These labs showcased Terraform's ability to automate scalable, secure, and highly available cloud infrastructure deployment.

After running the command:

```
terraform apply
```

go to the instance created

and connect to the instance and update it

```
sudo apt-get update
```

now we will install the mysql-client

```
sudo apt-get install mysql-client
```

Now we will connect to the mariadb using the hostname

```
public_ip = "3.82.44.238"
rds = "mariadb.cghaxz7pqg8a.us-east-1.rds.amazonaws.com:3306"
root@ip-172-31-17-1:~/workspace/terraform_for_aws_cloud_1/AWS_RDS#
```

```
mysql -u root -h mariadb.cghaxz7pqg8a.us-east-1.rds.amazonaws.com -p'mariadb141'
```

```
ubuntu@ip-10-0-1-28:~$ mysql -u root -h mariadb.cghaxz7pqg8a.us-east-1.rds.amazonaws.com -p'mariadb141'
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 26
Server version: 5.5.5-10.11.16-MariaDB-log managed by https://aws.amazon.com/rds/
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mariadb |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.01 sec)

mysql>
```

You can also see the database in the amazon console

Databases (1)

Filter by databases

DB id	Status	Role	Engine	Upgrad...	Region ...	Size	Recommendations	CPU	Current activity	Ma
mariadb	Available	Instance	MariaDB	SECOND	us-east-1a	db.t3.micro		5.53%	1 Connections	noi

Create IAM User using Terraform

To Begin with the Lab

So, here the most important thing to note is that after running the command

```
terraform apply
```

Don't run the command (given below)

```
terraform destroy
```

because terraform is third party application, this will create the problem in your AWS account.

It has successfully created the IAM Users.

IAM users (5) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in
adminuser1	/	1	-	-	-	-
adminuser2	/	1	-	-	-	-

adminuser1 Info

Summary

ARN: `arn:aws:iam::463646775279:user/adminuser1`

Created: June 19, 2026, 02:09 (UTC+05:30)

Console access: Disabled

Last console sign-in: -

Access key 1: [Create access key](#)

Permissions policies (1)

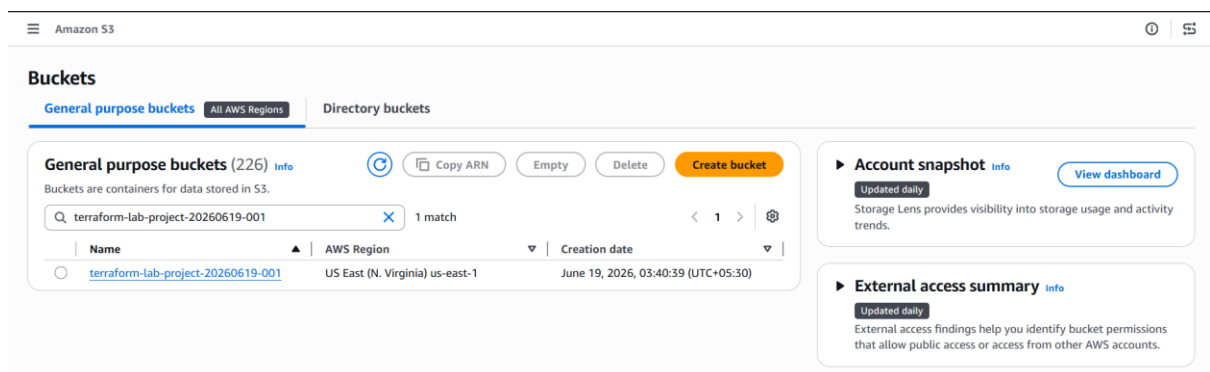
Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name	Type	Attached via
AdministratorAccess	AWS managed - job function	Group admingroup

AWS IAM Roles using Terraform

Created the bucket



Now connect to the created machine

Now go to the root user

```
sudo -s
```

Run the command

```
apt-get update
```

Now install the python and pip

```
apt-get install python3-pip python3-dev
```

Now download the AWS cli

```
apt install -y awscli
```

Now create a file

```
echo "Hi, This is my terraform class for AWS IAM" > myfile.txt
```

```
root@ip-172-31-27-208:/home/ubuntu# ls
root@ip-172-31-27-208:/home/ubuntu# @echo "Hi, This is my terraform class for AWS IAM" > myfile.txt
root@ip-172-31-27-208:/home/ubuntu# ls
myfile.txt
root@ip-172-31-27-208:/home/ubuntu#
```

Now upload the file in the bucket

```
aws s3 cp myfile.txt s3://terraform-lab-project-20260619-001
```

```
root@ip-172-31-27-208:/home/ubuntu# aws s3 cp myfile.txt s3://terraform-lab-project-20260619-001
upload: ./myfile.txt to s3://terraform-lab-project-20260619-001/myfile.txt
root@ip-172-31-27-208:/home/ubuntu#
```

Amazon S3 > Buckets > terraform-lab-project-20260619-001

terraform-lab-project-20260619-001 info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems - new | Access Points

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	myfile.txt	txt	June 19, 2026, 04:04:15 (UTC+05:30)	43.0 B	Standard

This is the role that is being used in the IAM roles

i-04d8e3c499741675f (custom_instance)

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details

IAM role s3-levelupbucket-role	Owner ID 463646775279	Launch time Fri Jun 19 2026 03:40:49 GMT+0530 (India Standard Time)
--	---	---

This is the permission being used in this role

IAM > Roles > s3-levelupbucket-role

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ Access reports

- Access Analyzer

Filter by Type: All types

☐	Policy name	Type	Attached entities
☐	s3-levelupmybucket-role-policy	Customer inline	0

s3-levelupmybucket-role-policy [Copy JSON](#) [Edit](#)

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Action": [
6         "s3:*"
7       ],
8       "Effect": "Allow",
9       "Resource": [
10        "arn:aws:s3:::terraform-lab-project-20260619-001",
11        "arn:aws:s3:::terraform-lab-project-20260619-001/*"
12      ]
13    }
14  ]
15 }
```

EC2 Instance Autoscaling using Terraform

To Begin with the Lab

Summary of the Lab

Open EC2 → Instances

You should see:

LevelUp Custom EC2 Instance

Notice:

Tag:

Name=LevelUp Custom EC2 Instance

This instance was created by the Auto Scaling Group.

Instances (1/2) [Info](#) Last updated 5 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set [Find Instance by attribute or tag \(case-sensitive\)](#)

[Instance state = running](#) [Clear filters](#)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DI
LevelUp Custom EC2 Instance	i-0cefa69c80ab57d9b	Running	t2.micro	2/2 checks passec	us-east-1f	ec2-100-24-1:

i-0cefa69c80ab57d9b (LevelUp Custom EC2 Instance)

Tags [Manage tags](#)

Key	Value
aws:ec2launc...	1
aws:autoscali...	levelup-autoscaling
Name	LevelUp Custom EC2 Instance
aws:ec2launc...	lt-03fc5e2388d81b27c

Verify CloudWatch Alarms

Open CloudWatch → Alarms

You should see:

levelup-cpu-alarm

levelup-cpu-alarm-scaledown

Expected initial state:

Run:

```
stress --cpu 2 --timeout 300
```

```
root@ip-172-31-64-168:~# stress --cpu 2 --timeout 300
stress: info: [1908] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
```

Meaning:

2 CPU workers

Run for 300 seconds (5 min)

This intentionally pushes CPU usage above 30%.

Watch CloudWatch

Open CloudWatch → Alarms → levelup-cpu-alarm

Wait 2–5 minutes.

CPU graph will increase.

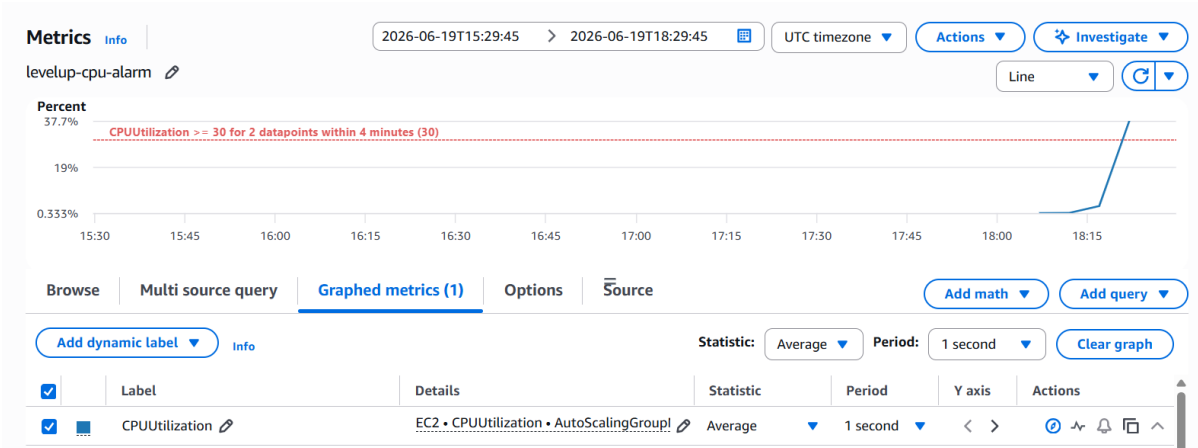
Eventually:

levelup-cpu-alarm

State = ALARM

Because:

CPU > 30%



Auto Scaling Triggered

Open EC2 → Auto Scaling Groups → levelup-autoscaling → Activity

You will see:

Launching a new EC2 instance

Desired Capacity changes:

1 → 2

EC2 > Auto Scaling groups

Auto Scaling groups (1) Info Last updated less than a minute ago

Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups

Name	Status - new	Launch template/configuration	Instances	Instance health - new	De
levelup-autoscaling	At desired capacity	levelup-launchtemplate2026061918093	2	2/2 Healthy	2

Verify New Instance

Open EC2 → Instances

Now you should see:

Instance #1 Running

Instance #2 Running

Instances (2) Info Last updated less than a minute ago

Connect Instance state Actions Launch instances

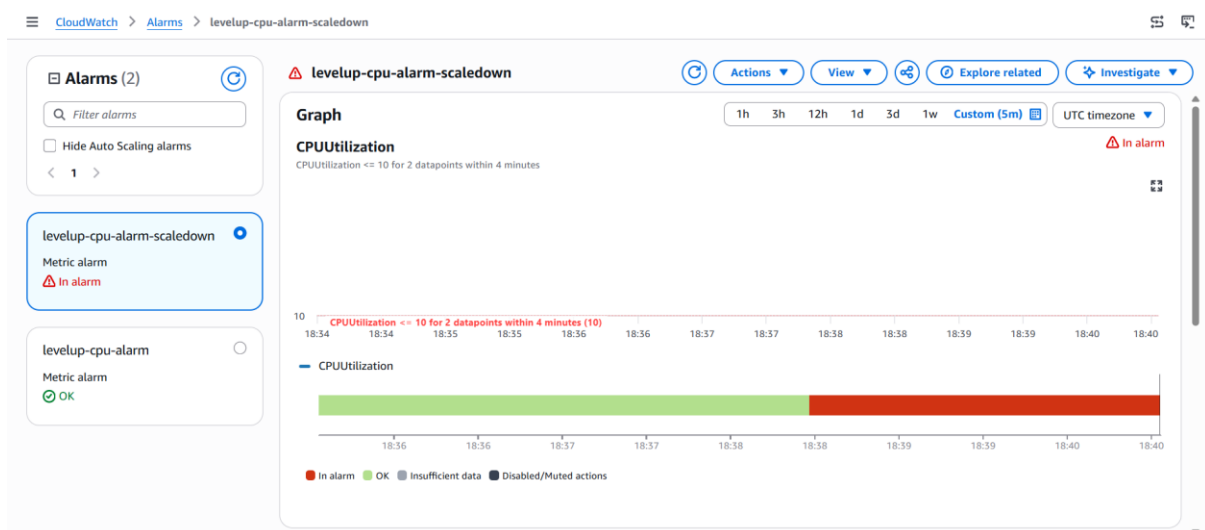
Saved filter sets Running (unsaved) Find Instance by attribute or tag (case-sensitive)

Instance state = running Key name = levelup_key Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DI
LevelUp Custom EC2 Instance	i-09926c6ccba1bb	Running	t2.micro	2/2 checks passed	us-east-1b	ec2-3-81-124
LevelUp Custom EC2 Instance	i-0cefa69c80ab57d9b	Running	t2.micro	2/2 checks passed	us-east-1f	ec2-100-24-1

Monitor the CloudWatch alarm levelup-cpu-alarm-scaledown; once CPU stays below 10% for several minutes, the alarm changes from OK → ALARM, triggering a scale-in event.

After 300 seconds, the stress test automatically stops, causing CPU utilization to drop back to normal levels.



In the Auto Scaling Group Activity, you'll see an EC2 instance being terminated and desired capacity reduced from 2 → 1; verify in EC2 → Instances that only 1 running instance remains.

Instances (2) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Running (unsaved)

Find Instance by attribute or tag (case-sensitive)

Key name = levelup_key

Clear filters

< 1 > ⚙

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DI
☐	LevelUp Custom EC2 Instance	i-09926c6ccbabea1bb	Terminated	t2.micro	-	us-east-1b	-
☐	LevelUp Custom EC2 Instance	i-0cefa69c80ab57d9b	Running	t2.micro	2/2 checks passed	us-east-1f	ec2-100-24-1:

Now run

```
terraform destroy
```

AWS Load Balancing using Terraform

To Begin with the Lab

Summary of the Lab

After the terraform apply

Use curl command with elb-DNS Name

```
curl < elb-dns-name >
```

It will show different IPs

You can copy and paste the DNS on the web browser and there also you can see the messages.